

Q. 1 (Probability gymnastics). Let A, B, C be any events.

- a. Let us write $B \setminus A$ as $B \cap A^c$. Now, since $(B \cap A^c) \cap (B \cap A) = \emptyset$, by the third axiom of probability measure we get:

$$\mathbb{P}(B) = \mathbb{P}((B \cap A^c) \cup (B \cap A)) = \mathbb{P}(B \cap A^c) + \mathbb{P}(B \cap A).$$

Since $A \subseteq B$, we notice that $A = B \cap A$ and therefore:

$$\mathbb{P}(B) = \mathbb{P}(B \cap A^c) + \mathbb{P}(A).$$

Moving $\mathbb{P}(A)$ from the right hand side of the equation to the left hand side we arrive at the required identity:

$$\mathbb{P}(B) - \mathbb{P}(A) = \mathbb{P}(B \cap A^c).$$

- b. Note that $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(A^c \cap B)$ since $A \cup B = A \cup (A^c \cap B)$ and $A \cap (A^c \cap B) = \emptyset$. The inequality then follows from $\mathbb{P}(A^c \cap B) \leq \mathbb{P}(B)$.
- c. By the definition of conditional probability,

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A|B \cap C)\mathbb{P}(B \cap C).$$

Applying the definition again for $\mathbb{P}(B \cap C)$ gives

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A|B \cap C)\mathbb{P}(B \cap C) = \mathbb{P}(A|B \cap C)\mathbb{P}(B|C)\mathbb{P}(C).$$

Q. 2 (Sherlock Holmes). Let A be the event that the fatal meeting was with her spouse, B be that with his business partner, C be that with his secretary, and D be that with the secret visitor. We have that $\mathbb{P}(A) = 0.15$, $\mathbb{P}(B) = 0.30$, $\mathbb{P}(C) = 0.50$, and $\mathbb{P}(D) = 0.05$.

Let S be the event that the man is shot. Sherlock deduces that $\mathbb{P}(S|A) = 0.05$, $\mathbb{P}(S|B) = 0.03$, $\mathbb{P}(S|C) = 0.01$, and $\mathbb{P}(S|D) = 0.05$. Using Bayes formula, the conditional probability that the man is killed by his wife is

$$\begin{aligned} \mathbb{P}(A|S) &= \frac{\mathbb{P}(A)\mathbb{P}(S|A)}{\mathbb{P}(A)\mathbb{P}(S|A) + \mathbb{P}(B)\mathbb{P}(S|B) + \mathbb{P}(C)\mathbb{P}(S|C) + \mathbb{P}(D)\mathbb{P}(S|D)} \\ &= \frac{0.15 \cdot 0.05}{0.15 \cdot 0.05 + 0.3 \cdot 0.03 + 0.5 \cdot 0.01 + 0.05 \cdot 0.05} = \frac{0.0075}{0.024} = 0.3125. \end{aligned}$$

Similarly, $\mathbb{P}(B|S) = 0.375$, $\mathbb{P}(C|S) \approx 0.2083$, and $\mathbb{P}(D|S) \approx 0.1042$. The business partner is the most likely to have been the murderer.

Note: there is no need to use the Law of Total Probability for the denominator in the Bayes formula as all the probabilities share the same denominator, so one can compare only the $\mathbb{P}(A \cap S)$, $\mathbb{P}(B \cap S)$, $\mathbb{P}(C \cap S)$, and $\mathbb{P}(D \cap S)$ terms.

Q. 3 (Help Chevalier de Méré). For each throw, the probability of getting a double six is

$$\mathbb{P}(\text{double six}) = \frac{1}{36}.$$

Then, by independence of the throws, the probability of not having a double six in the 24 throws is

$$\mathbb{P}(\text{no double six in 24 throws}) = \left(1 - \frac{1}{36}\right)^{24} \approx 0.5086.$$

Therefore the desired probability is

$$\mathbb{P}(\text{at least one double six}) = 1 - \left(1 - \frac{1}{36}\right)^{24} \approx 0.4914 < 0.5.$$

So it is not favorable for Chevalier to bet on a double six as he will win less than half the time.

Q. 4 (Bluephobia). Let us define T as the event that it takes 4 draws until there are only red balls in the urn. Note that the draws from the urn are not independent; indeed, the (conditional) probability of picking a blue ball in a draw changes according to the number of blue balls in the urn at the end of the previous draw.

- a. We are interested in computing $\mathbb{P}\{T\}$. Note that the process ends at the end of the fourth draw if and only if the last blue ball in the urn is picked in the fourth draw. Formally,

$$\mathbb{P}\{T\} = \mathbb{P}\{BRRB\} + \mathbb{P}\{RBRB\} + \mathbb{P}\{RRBB\}$$

since there are three possible outcomes will let the ball draw finish at the end of fourth draw.

To compute the first term on the right hand side, we have

$$\begin{aligned} \mathbb{P}\{BRRB\} &= \mathbb{P}\{B\}\mathbb{P}\{BR|B\}\mathbb{P}\{BRR|BR\}\mathbb{P}\{BRRB|BRR\} \\ &= \frac{2}{6} \cdot \frac{4}{5} \cdot \frac{4}{5} \cdot \frac{1}{5}. \end{aligned}$$

Similarly, for the second and third terms we have:

$$\begin{aligned} \mathbb{P}\{RBRB\} &= \frac{4}{6} \cdot \frac{2}{6} \cdot \frac{4}{5} \cdot \frac{1}{5}, \\ \mathbb{P}\{RRBB\} &= \frac{4}{6} \cdot \frac{4}{6} \cdot \frac{2}{6} \cdot \frac{1}{5}. \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbb{P}\{T\} &= \mathbb{P}\{BRRB\} + \mathbb{P}\{RBRB\} + \mathbb{P}\{RRBB\} \\ &= \left(\frac{2}{6} \cdot \frac{4}{5} \cdot \frac{4}{5} + \frac{4}{6} \cdot \frac{2}{6} \cdot \frac{4}{5} + \frac{4}{6} \cdot \frac{4}{6} \cdot \frac{2}{6}\right) \cdot \frac{1}{5} \\ &= \left(\frac{1}{5 \cdot 5} + \frac{1}{6 \cdot 5} + \frac{1}{6 \cdot 6}\right) \cdot \frac{2 \cdot 4 \cdot 4}{6 \cdot 5} \\ &= \frac{36 + 30 + 25}{25 \cdot 36} \cdot \frac{32}{30} \\ &= \frac{91 \cdot 4}{25 \cdot 9 \cdot 15} \\ &= \frac{364}{3375} \approx 0.108. \end{aligned}$$

- b. Let A be the event that a blue ball is picked in the third draw. We have

$$\mathbb{P}\{A|T\} = \frac{\mathbb{P}\{A \cap T\}}{\mathbb{P}\{T\}},$$

Since the event A is simply represented as $RRBB$, in which the first two draws yield red balls and the third and fourth draw yield blue balls, we have:

$$\mathbb{P}(A|T) = \frac{\mathbb{P}(A \cap T)}{\mathbb{P}\{T\}} = \frac{\mathbb{P}\{RRBB\}}{\mathbb{P}\{BRRB\} + \mathbb{P}\{RBRB\} + \mathbb{P}\{RRBB\}} \approx 0.275.$$

Q. 5 (Alice and Bob have fun with coins). Let us denote by HH, HT, TH, TT the result of a pair of flips, where the first letter denotes the result of Alice's flip and the second letter denotes the result of Bob's flip. Let X_k be the result of the k th pair of flips.¹ The random variable X_k takes one of the values in $\{HH, HT, TH, TT\}$. Let A_k be the event that Alice wins in the k -th round.

- (a) A_1 is the event that Alice wins the game in the first round. It is clear that we may write

$$A_1 = \{X_1 = HT\},$$

and

$$\mathbb{P}(A_1) = p_A q_B = p_A(1 - p_B).$$

- (b) A_2 is the event that Alice wins the game in the second round. For Alice to win in the second round, it is necessary and sufficient that the first round ends in a tie (either HH or TT) and the second game comes up HT . Formally, we may write

$$A_2 = \{X_1 \in \{HH, TT\}\} \cap \{X_2 = HT\}.$$

Since the coins are performed independently,

$$\mathbb{P}(A_2) = (p_A p_B + q_A q_B) p_A q_B = (2p_A p_B - p_A - p_B + 1) p_A(1 - p_B).$$

- (c) A_k is the event that Alice wins the game in the k -th round. For Alice to win at the k -th pair, it is necessary and sufficient that the first $k - 1$ pairs result in a tie (either HH or TT) and the n th pair comes up HT . Formally, we may write

$$A_k = \{X_1 \in \{HH, TT\}\} \cap \dots \cap \{X_{k-1} \in \{HH, TT\}\} \cap \{X_k = HT\}.$$

Since the coin flips are performed independently,

$$\begin{aligned} \mathbb{P}(A_k) &= \mathbb{P}\{X_1 \in \{HH, TT\}\} \cdot \dots \cdot \mathbb{P}\{X_{k-1} \in \{HH, TT\}\} \cdot \mathbb{P}\{X_k = HT\} \\ &= (p_A p_B + q_A q_B) \cdot \dots \cdot (p_A p_B + q_A q_B) \cdot p_A q_B \\ &= (p_A p_B + q_A q_B)^{k-1} p_A q_B \\ &= (2p_A p_B - p_A - p_B + 1)^{k-1} p_A(1 - p_B). \end{aligned}$$

- (d) Let A be the event that Alice eventually wins the game. For each $k \geq 1$, let A_k be the event that Alice wins the game at the k -th pair of flip. Clearly, $A_1, A_2, \dots$ are mutually exclusive events. Hence,

$$\mathbb{P}(A) = \sum_{k=1}^{\infty} \mathbb{P}(A_k).$$

Since the coin flips are performed independently, we already know that

$$\mathbb{P}(A_k) = (p_A p_B + q_A q_B)^{k-1} p_A q_B.$$

Therefore,

$$\begin{aligned} \mathbb{P}(A) &= \sum_{k=1}^{\infty} \mathbb{P}(A_k) = \sum_{k=1}^{\infty} (p_A p_B + q_A q_B)^{k-1} p_A q_B \\ &= \frac{p_A q_B}{1 - (p_A p_B + q_A q_B)} = \frac{p_A(1 - p_B)}{p_A + p_B - 2p_A p_B}. \end{aligned}$$

¹This definition of X_k makes sense as long as the game has not ended by the k -th flip. Otherwise, if the k -th pair of flips is not to be performed for some outcome $\omega \in \Omega$, then we may define, for instance, $X_k(\omega) = \Delta$ for definiteness.

In this computation, note that we use the geometric series formula

$$1 + r + r^2 + \dots = \sum_{k=0}^{\infty} r^k = \sum_{k=1}^{\infty} r^{k-1} = \frac{1}{1-r}$$

for a real number r with $|r| < 1$. In the above computation, we take

$$r = p_A p_B + q_A q_B = 1 - p_A - p_B + 2p_A p_B.$$

- (e) For the game to be fair, Alice and Bob must have equal chances of winning the game. Let B be the event that Bob wins the game. Switching the roles of the first and second flips in a pair and the roles of p_A and p_B in part (a) yields

$$\mathbb{P}(B) = \frac{p_B(1-p_A)}{p_B + p_A - 2p_B p_A}.$$

Indeed,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) = \frac{p_A(1-p_B) + p_B(1-p_A)}{p_A + p_B - 2p_A p_B} = 1.$$

In other words, the game ends after some finite (but random) number of pairs of flips with probability one, or equivalently, the probability that the game continues indefinitely is zero. (This is an instance of a general principle in the study of Bernoulli processes which we will discuss in the lectures.)

Therefore, the game is fair if and only if

$$\mathbb{P}(A) = \frac{p_A(1-p_B)}{p_A + p_B - 2p_A p_B} = \frac{1}{2},$$

which is equivalent to

$$p_A = p_B \in (0, 1).$$